

Lecture 19

DC MOSFET Analysis

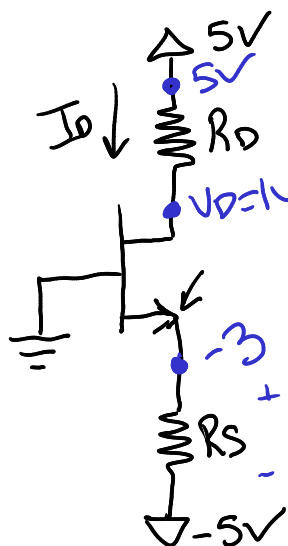
Choose R_D and R_S such that NFET

$$V_D = 1V \text{ and } I_D = 400\mu A$$

Assume $V_{TN} = 2V$

$$k'_n = 20 \frac{\mu A}{V^2} \quad \frac{W}{L} = \frac{400\mu m}{10\mu m}$$

Determine operating region



Is the MOSFET on? Assume it is off, $I_D = 0$

$$V_{RS} = 0V \Rightarrow V_S = -5V$$

$$\text{What is } V_{GS}? \Rightarrow V_{GS} = 0 - (-5) = 5V$$

$V_{GS} > V_{TN}$ MOSFET is ON!

* Assume it is in Saturation

$$\therefore V_{DS} \geq V_{GS} - V_{TN} \Rightarrow V_D - V_S \geq V_G - V_S - V_{TN}$$

$$\Rightarrow V_D \geq V_G - V_{TN}$$

Boundary Condition

We wanted $V_D = 1V$

Is $1V > 0 - 2V \Rightarrow$ Yes! It is SAT. by inspection

$$R_D = \frac{5-1}{400\mu A} = \boxed{10k\Omega}$$

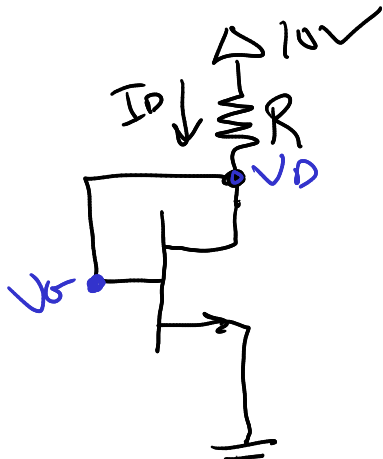
$$\text{For } R_S: I_D = 400\mu A = \frac{1}{2} k'_n \frac{W}{L} (V_{GS} - V_{TN})^2$$

$$\Rightarrow 400 \times 10^{-6} = (10 \times 10^{-6}) (40) (0 - V_S - 2)^2$$

Quadratic Solve: $V_S = -3V$ or $V_S = -1V$
Which one? $V_{GS} = 3V$ or $V_{GS} = 1V$

$$R_S = \frac{-3+5}{400\mu A} = \boxed{5k\Omega}$$

2)



Choose R to set $I_D = 400\mu A$

What will be $V_D = ?$

$V_D = V_G$ by inspection

Assume $k_n' = \frac{20\mu A}{V^2}$, $\frac{W}{L} = \frac{100\mu m}{10\mu m}$, $V_{TN} = 2V$

$\therefore V_D > V_G - V_{TN}$ by inspection

If the MOSFET is ON, must be sat.

* Is it ON? Assume it's off and $I_D = 0 A$

$V_R = 0V \Rightarrow V_D = 10V$ and $V_G = 10V$

$V_{GS} = 10 > V_{TN}$ It is ON

$$I_D = 400\mu A = \frac{1}{2} k_n' \frac{W}{L} (V_G - V_S - V_{TN})^2$$

$$\Rightarrow 400\mu A = 160 \times 10^{-6} (V_G - 0 - 2)^2$$

Solve $V_G = 0V$ or $V_G = 4V$ ✓
 $V_{GS} = 0V$ $V_{GS} = 4V$

$$R_D = \frac{10 - 4}{400\mu A} = 15k\Omega$$

* This is a diode connected MOSFET
 MOSFET = drain connected gate

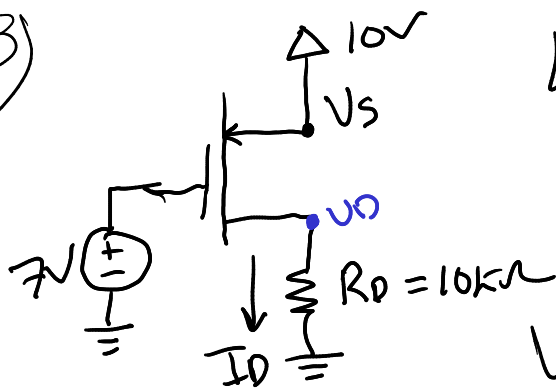
(1) If the MOSFET is on, we know it's SAT.

(2) Similar Diode, only conducting when $V_D > V_{TN}$

(3) Essential to current mirrors / set reference current

(4) Ideal for IC design

3)



Let $K'_p = 10 \frac{\mu A}{V^2} \frac{W}{L} = 20$

$V_{TP} = -1V$

Find V_D and I_D

$V_{SG} = 10 - 7 = 3$

Is $3 > |V_{TP}|$ Yes it is on

Assume Saturation, $I_D = \frac{1}{2} K'_p \frac{W}{L} (V_{SG} - |V_{TP}|)^2$

$100 \times 10^{-6} (3 - 1)^2 = 400 \mu A$

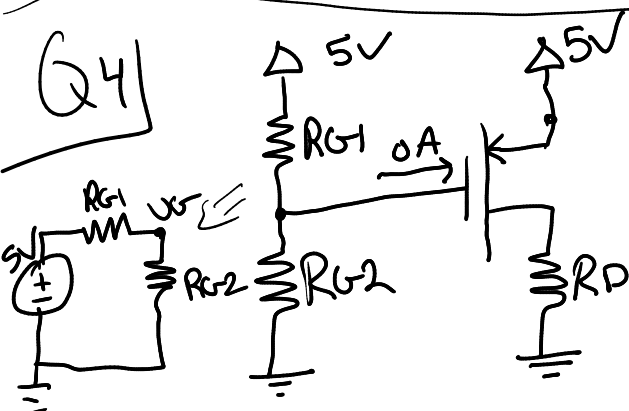
$V_D = I_D R_D = 10 \times 10^3 (400 \times 10^{-6}) = 4V$

Is $V_{SD} \geq V_{SG} - |V_{TP}|$?

$6 \geq 3 - 1$ Yes it is SAT.



Q4)



Let $K_P = 1 \frac{mA}{V^2}$, $V_{TP} = -1V$

Design circuit so that

$I_D = 0.5mA$

$V_D = 3V$

PFE T must be Saturated

$R_D = \frac{3 - 0}{0.5mA} = 6k\Omega$

$I_D = \frac{1}{2} K_P (V_{SG} - |V_{TP}|)^2$

$0.5mA = 0.5 \frac{mA}{V^2} (5 - V_G - 1)^2$

$\Rightarrow V_G = 5V$ or $V_G = 3V$

$V_{SG} = 0V$

$V_{SG} = 2V$ ✓✓

$3 = \frac{R_{G2}}{R_{G1} + R_{G2}} (5)$ 2 unknowns
1 equation

Infinite # of combos to make V_G

Pick one! $R_{G2} = 300k\Omega$
 $R_{G2} = 30k\Omega$

$\Rightarrow R_{G1} = 200k\Omega$
 $\Rightarrow R_{G1} = 20k\Omega$

Larger Resistor $\rightarrow$ lower quiescent current, less loss

Smaller $\rightarrow$ less area on an IC
Save you money